Code No. 3419

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) IV-Semester (MAIN) Examinations Jun/Jul-2026

DATA ANALYTICS USING R

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	What is the output of the given R statements: **a = c(10, 20, 30, 40, 50, 60)** **n = array(a, dim = c(2, 2, 3))** **print(n)**	2	3	1
b)	Write the output of the R program a <- c(11,12,34,42,50) b <- c("R", "Is", "Fun!","Let's","Learn") c <- c(TRUE,FALSE,TRUE,TRUE,FALSE) data.frame(a,b,c) my_frame <- data.frame(a,b,c) print(my_frame)	2	4	2
c)	Define clustering and its purpose.	2	2	3
d)	Write a R program to print the sum of series (use iterative statements only) (Read input x and n) $S = 1 + x^2/2! + x^3/3! + \ldots\ldots + x^n/n!$	2	1	4
e)	Differentiate between linear regression and logistic regression.	2	2	5
f)	Write R code to read a CSV file and display the first 10 rows.	2	3	1
2. a)	Write about the different data types in R with examples. Also explain type coercion.	6	4	1
b)	Consider titanic dataset to find the missing values and duplicates. Explain step-by-step procedure to find the missing values and duplicates. Write the code in R.	6	3	1
3. a)	Analyze the concept of linear regression, including model equation and interpretation of coefficients.	6	4	2
b)	Name any two plotting functions in R and give example for each.	3	5	2

(P.T.O.)

	c)	Write a program to create a matrix using cbind() and rbind() functions.	3	2	1
4.	a)	Describe entropy and information gain. How are they used to select attributes in decision trees?	6	4	3
	b)	Investigate the working of k-means algorithm along with the challenges such as choosing the value of k.	6	4	3
5.	a)	Elaborate the text mining process with all steps involved.	6	2	4
	b)	Write function named as Factorial() using R language to find the factorial of a given number.	3	4	4
	c)	Write the syntax of while loop and for loop in R? Give example for each.	3	2	1
6.	a)	Elucidate different exponential smoothing techniques: simple, Holt, and Holt-Winters.	6	2	5
	b)	Explain all R data structures with examples and use cases.	6	3	5
7.	a)	Construct vectors in R, including sequence vectors, vector access, and vector recycling with examples.	6	6	1
	b)	Evaluate association rule mining and its key concepts: frequent itemsets, rules, and evaluation metrics.	6	5	3

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